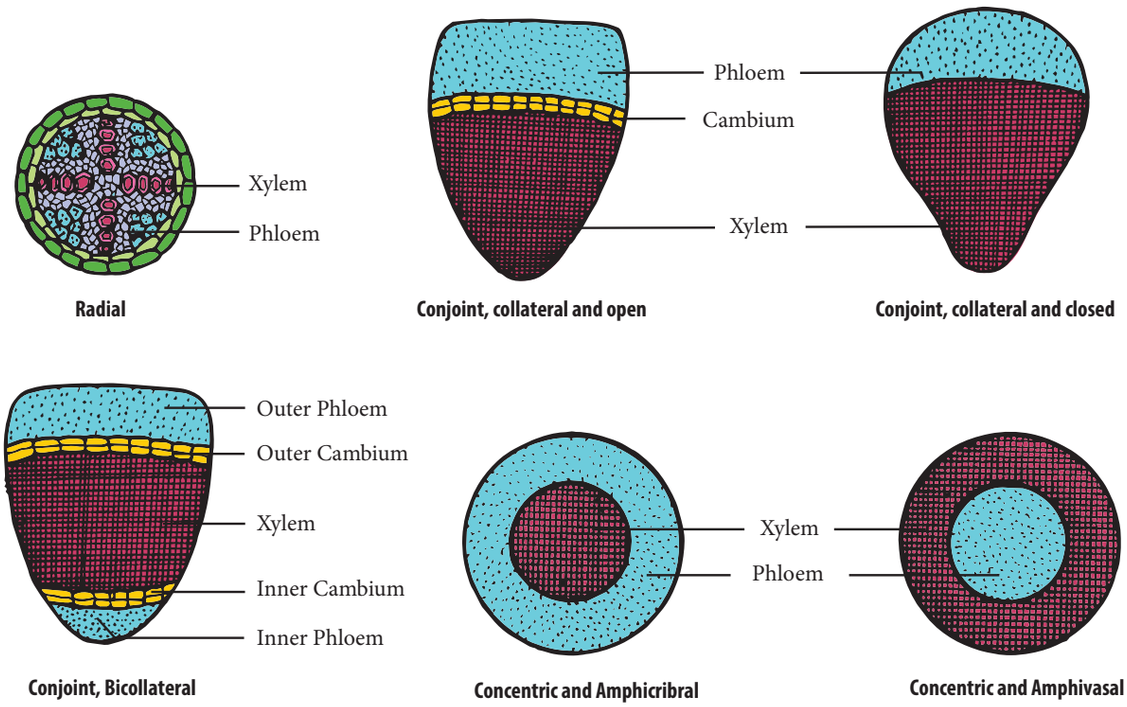


Figure 12.1 Types of vascular bundle

12.3 Internal Structure of Dicot Root (Bean)

A thin transverse section of dicot root shows the following structures.

(i) Epiblema: It is the outermost layer. Cuticle and stomata are absent. Unicellular root hairs are present. It is also known as **Rhizodermis** or **Piliferous layer**.

(ii) Cortex: It is a multilayered large zone made of thin-walled parenchymatous cells with intercellular spaces. It stores food and water.

(iii) Endodermis: It is the innermost layer of cortex. The cells are barrel - shaped, closely packed, and show band like thickenings on their radial and inner tangential walls called **casparian strips**. But these casparian strips are absent in the endodermis cells which are located opposite the protoxylem these thin walled cells without casparian strips are called passage cell. It helps in the movement of water and dissolved salts from cortex into xylem.

(iv) Stele: All tissues inner to endodermis constitute stele. It includes pericycle and vascular bundle.

(a) Pericycle: Inner to endodermis lies a single layer of pericycle. It is the site of origin of lateral roots.

(b) Vascular bundle: It is radial. Xylem is **exarch** and **tetrach**. The tissue present between xylem and phloem is called conjunctive tissue. In dicot root, it is made up of parenchyma.

(c) Pith: Young root contains pith whereas in old root pith is absent.

12.4 Internal Structure of Monocot Root (Maize)

A thin transverse section of monocot root, shows the following characteristic features.

i. Epiblema or Rhizodermis: It is the outermost layer of the root, and is made up of single layer of thin walled, parenchymatous cell. Stomata and cuticle are absent. The root hair helps in absorption of water and minerals from the soil. This layer also protects the inner tissues.

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